

numpy

May 19, 2026

```
[ ]: # Numpy Crash Course
```

```
[14]: import numpy as np
```

```
[15]: np.__version__
```

```
[15]: '2.3.5'
```

```
[16]: import sys  
      sys.version
```

```
[16]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929  
      64 bit (AMD64)]'
```

```
[ ]: ## Creating Arrays
```

```
[17]: my_list = [0,1,2,3,4,5]  
      my_list
```

```
[17]: [0, 1, 2, 3, 4, 5]
```

```
[18]: type(my_list)
```

```
[18]: list
```

```
[19]: arr = np.array(my_list)
```

```
[20]: arr
```

```
[20]: array([0, 1, 2, 3, 4, 5])
```

```
[21]: type(arr)
```

```
[21]: numpy.ndarray
```

```
[22]: type(my_list)
```

```
[22]: list
```

```
[23]: # numpy arrange
```

```
[24]: np. # we learn important function
```

```
Cell In[24], line 1
    np. # we learn important function
    ^
SyntaxError: invalid syntax
```

```
[25]: np.arange(15)
```

```
[25]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14])
```

```
[26]: np.arange(3.0)
```

```
[26]: array([0., 1., 2.])
```

```
[27]: np.arange(10)
```

```
[27]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
[28]: np.arange(0, 5)
```

```
[28]: array([0, 1, 2, 3, 4])
```

```
[29]: np.arange(10,20)
```

```
[29]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])
```

```
[30]: np.arange(20,10) # 1st arg < 2nd arg
```

```
[30]: array([], dtype=int64)
```

```
[31]: np.arange(-20,10)
```

```
[31]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
          -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
           6,  7,  8,  9])
```

```
[32]: np.arange(-16,10)
```

```
[32]: array([-16, -15, -14, -13, -12, -11, -10, -9, -8, -7, -6, -5, -4,
          -3, -2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  9])
```

```
[33]: np.arange(-20,-10)
```

```
[33]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])
```

```
[34]: np.arange(30,20)  # 1st arg always be < then 2nd arg
```

```
[34]: array([], dtype=int64)
```

```
[35]: np.arange()
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[35], line 1  
----> 1 np.arange()  
  
TypeError: arange() requires stop to be specified.
```

```
[36]: np.arange(10,30,5)  # 10- starting from 30- end point 5 - step count
```

```
[36]: array([10, 15, 20, 25])
```

```
[37]: np.arange(0,10,3)
```

```
[37]: array([0, 3, 6, 9])
```

```
[38]: np.arange(10,30,5,8)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[38], line 1  
----> 1 np.arange(10,30,5,8)  
  
TypeError: Cannot interpret '8' as a data type
```

```
[39]: np.zeros(3) # parameter tuning
```

```
[39]: array([0., 0., 0.])
```

```
[40]: np.zeros(5, dtype=int) # hyperparameter tuning
```

```
[40]: array([0, 0, 0, 0, 0])
```

```
[41]: np.zeros((2,2), dtype=int)
```

```
[41]: array([[0, 0],  
          [0, 0]])
```

```
[42]: zero = np.zeros([2,2])
      print(zero)
      print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
<class 'numpy.ndarray'>
```

```
[43]: zero = np.zeros([2,2])
      print(zero)

      print('####')

      print(type(zero))
```

```
[[0. 0.]
 [0. 0.]]
####
<class 'numpy.ndarray'>
```

```
[44]: np.zeros((2,10))
```

```
[44]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
[45]: np.zeros((2,2))
```

```
[45]: array([[0., 0.],
            [0., 0.]])
```

```
[46]: np.zeros((3,3))
```

```
[46]: array([[0., 0., 0.],
            [0., 0., 0.],
            [0., 0., 0.]])
```

```
[47]: np.zeros((10,30))
```

```
[47]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,  
0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,  
0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,  
0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,  
0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
[0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.,  
0., 0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
[48]: np.zeros((5,10)) # by default large -- will give row & 2nd arg - columns
```

```
[48]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],
            [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
[49]: n = (6,7)
      n1 = (6,8)
      print(np.zeros(n1)) # parameter tuning
```

```
[0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0. 0.]
```

```
[50]: print(np.zeros(n,dtype=int)) ## hyperparameter tunni
```

[illegible]

[51]: `n`

[51] : (6, 7)

```
[52]: n1
```

[52] : (6, 8)

```
[53]: print(np.zeros(n1))
```

```
[[0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0.]]
```

```
[54]: # numpy with ones
```

```
[55]: np.ones(3)
```

```
[55]: array([1., 1., 1.])
```

```
[56]: np.ones(4, dtype=int)
```

```
[56]: array([1, 1, 1, 1])
```

```
[57]: np.ones(4)
```

```
[57]: array([1., 1., 1., 1.])
```

```
[58]: n
```

```
[58]: (6, 7)
```

```
[59]: np.ones(n)
```

```
[59]: array([[1., 1., 1., 1., 1., 1., 1.],
            [1., 1., 1., 1., 1., 1., 1.],
            [1., 1., 1., 1., 1., 1., 1.],
            [1., 1., 1., 1., 1., 1., 1.],
            [1., 1., 1., 1., 1., 1., 1.],
            [1., 1., 1., 1., 1., 1., 1.]])
```

```
[60]: np.ones((5,4),dtype=int) # by default 5- rows & 4 - columns
```

```
[60]: array([[1, 1, 1, 1],
            [1, 1, 1, 1],
            [1, 1, 1, 1],
            [1, 1, 1, 1],
            [1, 1, 1, 1]])
```

```
[61]: np.
```

```
Cell In[61], line 1
    np.
      ^
SyntaxError: invalid syntax
```

```
[62]: np.twos((2,3))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[62], line 1
----> 1 np.twos((2,3))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

```
[63]: np.three(2,3)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[63], line 1
----> 1 np.three(2,3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'
```

```
[64]: np.ones(2)
```

```
[64]: array([1., 1.])
```

```
[65]: np.ones((2,4))
```

```
[65]: array([[1., 1., 1., 1.],
           [1., 1., 1., 1.]])
```

```
[66]: np.ones((6,10), dtype = int)
```

```
[66]: array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])
```

```
[67]: np.twos((2,4))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[67], line 1
----> 1 np.twos((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

```
[68]: np.
```

```
Cell In[68], line 1
    np.
    ~
SyntaxError: invalid syntax
```

```
[69]: np.three((2,4))
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[69], line 1
----> 1 np.three((2,4))

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'three'
```

```
[70]: range(5)
```



```
[70]: range(0, 5)
```

```
[71]: r = range(5)
      r
```

```
[71]: range(0, 5)
```

```
[72]: for i in r:
      print(i)
```

```
0
1
2
3
4
```

```
[73]: list(range(5))
```

```
[73]: [0, 1, 2, 3, 4]
```

```
[74]: range(1,10)
```

```
[74]: range(1, 10)
```

```
[75]: list(range(1,10))
```

```
[75]: [1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
[76]: list(range(1,10,3))
```

```
[76]: [1, 4, 7]
```

```
[77]: y = list(range(12))
      y
```

```
[77]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
```

```
[78]: # Random value
```

```
[79]: rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[79], line 1
----> 1 rand(3,2)

NameError: name 'rand' is not defined
```

```
[80]: rand(3,2)
      random.rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[80], line 1
----> 1 rand(3,2)
      2 random.rand(3,2)

NameError: name 'rand' is not defined
```

```
[81]: np.random.rand(5)
```

```
[81]: array([0.71665763, 0.79156774, 0.09054909, 0.72191213, 0.51658535])
```

```
[82]: np.rand(4)
```

```
-----
AttributeError                            Traceback (most recent call last)
Cell In[82], line 1
----> 1 np.rand(4)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'rand'
```

```
[91]: np.random.rand(2,4)
```

```
[91]: array([[0.64538749, 0.83053931, 0.10923398, 0.60533534],
            [0.20449444, 0.8687024 , 0.96161827, 0.19648921]])
```

```
[92]: np.random.randint(2,4)
```

```
[92]: 2
```

```
[85]: np.random.randint(2,20) # 2nd argument is exclusive
```

```
[85]: 14
```

```
[86]: np.random.randint(0,1)
```

```
[86]: 0
```

```
[87]: np.random.randint(10,20,5)
```

```
[87]: array([15, 18, 19, 18, 12], dtype=int32)
```

```
[88]: np.random.randint(1,6,4)
```

```
[88]: array([4, 3, 2, 2], dtype=int32)
```

```
[89]: np.random.rand(3)
```

```
[89]: array([0.37246884, 0.73614186, 0.47059295])
```

```
[90]: np.random.randint(1)
```

```
[90]: 0
```

```
[93]: np.random.randint(30,20,10)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[93], line 1
----> 1 np.random.randint(30,20,10)

File numpy/random/mtrand.pyx:794, in numpy.random.mtrand.RandomState.randint()

File numpy/random/_bounded_integers.pyx:1425, in numpy.random._bounded_integers
↳ _rand_int32()

ValueError: low >= high
```

```
[94]: np.random.randint(-30,20,10)
```

```
[94]: array([-30,  -3,  15, -29, -15, -18, -17,   3,  12, -16], dtype=int32)
```

```
[95]: np.random.randint(20,30,10)
```

```
[95]: array([20, 28, 20, 22, 23, 25, 20, 25, 20, 27], dtype=int32)
```

```
[96]: np.random.randint(5,9)  #GET THE VALUE <=1 & >=5
```

```
[96]: 6
```

```
[97]: np.random.randint(10,21,3)
```

```
[97]: array([10, 18, 20], dtype=int32)
```

```
[98]: np.random.randint(1,12,10)
```

```
[98]: array([11,  3,  5, 11,  3,  6,  8,  5,  6,  7], dtype=int32)
```

```
[99]: np.random.randint(10,40,(10,10)) #generre the element 10 -30 with 4*4 mtri
```

```
[99]: array([[38, 21, 30, 39, 19, 22, 39, 21, 30, 31],
            [33, 22, 10, 34, 29, 21, 17, 15, 39, 22],
            [38, 27, 25, 15, 14, 39, 23, 31, 25, 15],
            [13, 25, 36, 27, 25, 11, 35, 35, 18, 19],
            [33, 37, 20, 36, 18, 18, 17, 35, 25, 25],
            [39, 38, 26, 33, 37, 16, 20, 13, 38, 16],
            [10, 22, 20, 19, 29, 33, 37, 26, 33, 26],
            [33, 39, 38, 24, 39, 17, 37, 23, 26, 18],
            [11, 39, 34, 35, 29, 29, 33, 27, 13, 16],
            [29, 25, 12, 13, 11, 24, 24, 14, 19, 39]], dtype=int32)
```

```
[100]: np.random.randint(1,100,(12,12)) #generre the element 10 -30 with 4*4 mtri
```

```
[100]: array([[67, 85, 10, 42, 96, 60, 95, 42, 96, 79, 61, 86],
            [89, 94, 63, 64, 54, 86, 81, 17, 63, 29, 33, 26],
            [34, 95, 51, 13, 32, 92, 59, 88, 73, 10, 27,  1],
            [39, 74, 67, 96, 80, 92, 25, 70, 18, 28, 11, 54],
            [50, 61, 54, 80, 55, 15, 24, 97, 39, 38, 39,  6],
            [95, 44, 21, 33, 64, 91, 38, 57, 54,  8, 49, 68],
            [64, 39, 14, 63, 71, 29, 91, 19, 78, 49,  4, 11],
            [90, 18, 29,  6,  9, 19, 34, 95, 50, 60, 99, 86],
            [33, 41, 37, 25, 52, 53, 43, 27, 17, 52, 79,  1],
            [90, 56, 43, 69, 47, 17, 61, 19, 26, 75, 13,  4],
            [55, 64, 12,  3, 40, 85, 60, 41, 63, 65, 29, 75],
            [75, 24,  9, 75, 32, 51, 41, 52, 23, 92, 17, 35]], dtype=int32)
```

```
[101]: np.arange(1,13).reshape(3,4)
```

```
[101]: array([[ 1,  2,  3,  4],
            [ 5,  6,  7,  8],
            [ 9, 10, 11, 12]])
```

```
[102]: np.arange(1,13).reshape(12, 1)
```

```
[102]: array([[ 1],
            [ 2],
            [ 3],
            [ 4],
            [ 5],
            [ 6],
            [ 7],
```

```
[ 8],  
[ 9],  
[10],  
[11],  
[12]])
```

```
[103]: b = np.random.randint(10,20,(5,4))  
b
```

```
[103]: array([[10, 15, 11, 15],  
             [11, 18, 17, 10],  
             [10, 17, 12, 11],  
             [11, 14, 19, 18],  
             [11, 18, 16, 17]], dtype=int32)
```

```
[104]: type(b)
```

```
[104]: numpy.ndarray
```

```
[105]: b
```

```
[105]: array([[10, 15, 11, 15],  
             [11, 18, 17, 10],  
             [10, 17, 12, 11],  
             [11, 14, 19, 18],  
             [11, 18, 16, 17]], dtype=int32)
```

```
[106]: b[:]
```

```
[106]: array([[10, 15, 11, 15],  
             [11, 18, 17, 10],  
             [10, 17, 12, 11],  
             [11, 14, 19, 18],  
             [11, 18, 16, 17]], dtype=int32)
```

```
[107]: b[1:3]
```

```
[107]: array([[11, 18, 17, 10],  
             [10, 17, 12, 11]], dtype=int32)
```

```
[108]: b
```

```
[108]: array([[10, 15, 11, 15],  
             [11, 18, 17, 10],  
             [10, 17, 12, 11],  
             [11, 14, 19, 18],  
             [11, 18, 16, 17]], dtype=int32)
```

```

[109]: b[1,2]

[109]: np.int32(17)

[110]: b

[110]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)

[111]: b[1,3]

[111]: np.int32(10)

[112]: b[1,-1]

[112]: np.int32(10)

[113]: b

[113]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)

[114]: b[2:3]

[114]: array([[10, 17, 12, 11]], dtype=int32)

[115]: b

[115]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)

[116]: b[0:-2]

[116]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11]], dtype=int32)

[117]: b

```

```
[117]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)
```

```
[118]: b[-5,-3]
```

```
[118]: np.int32(15)
```

```
[119]: b
```

```
[119]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)
```

```
[120]: b[-4,2]
```

```
[120]: np.int32(17)
```

```
[121]: np.random.randint(10,20,(4,4))
```

```
[121]: array([[18, 18, 14, 13],
             [14, 14, 10, 18],
             [10, 12, 18, 15],
             [13, 11, 19, 19]], dtype=int32)
```

```
[122]: b
```

```
[122]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
             [11, 18, 16, 17]], dtype=int32)
```

```
[123]: b[-4,-2]
```

```
[123]: np.int32(17)
```

```
[124]: b
```

```
[124]: array([[10, 15, 11, 15],
             [11, 18, 17, 10],
             [10, 17, 12, 11],
             [11, 14, 19, 18],
```

```
[11, 18, 16, 17]], dtype=int32)
```

```
[125]: b[-4:2]
```

```
[125]: array([[11, 18, 17, 10]], dtype=int32)
```

```
[126]: b[:]
```

```
[126]: array([[10, 15, 11, 15],  
            [11, 18, 17, 10],  
            [10, 17, 12, 11],  
            [11, 14, 19, 18],  
            [11, 18, 16, 17]], dtype=int32)
```

```
[127]: ## Operations
```

```
[128]: a = np.random.randint(10,20,10)  
a
```

```
[128]: array([11, 17, 15, 12, 11, 11, 18, 18, 17, 19], dtype=int32)
```

```
[129]: id(a)
```

```
[129]: 2123378752272
```

```
[130]: arr
```

```
[130]: array([0, 1, 2, 3, 4, 5])
```

```
[131]: arr2 = np.random.randint(0,100,(10,10))
```

```
[132]: arr2
```

```
[132]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],  
            [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],  
            [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],  
            [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],  
            [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],  
            [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],  
            [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],  
            [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],  
            [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],  
            [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[133]: arr
```

```
[133]: array([0, 1, 2, 3, 4, 5])
```



```
[134]: arr[:]
```

```
[134]: array([0, 1, 2, 3, 4, 5])
```

```
[135]: arr
```

```
[135]: array([0, 1, 2, 3, 4, 5])
```

```
[136]: arr[:4]
```

```
[136]: array([0, 1, 2, 3])
```

```
[137]: arr2[:]
```

```
[137]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
             [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
             [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
             [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
             [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
             [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[138]: arr2[0:5]
```

```
[138]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21]], dtype=int32)
```

```
[139]: arr2
```

```
[139]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
             [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
             [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
             [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
             [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
             [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[140]: arr2[1,4]
```

```
[140]: np.int32(49)
```

```
[141]: arr2
```

```
[141]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
             [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
             [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
             [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
             [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
             [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[142]: arr2[-5,5]
```

```
[142]: np.int32(49)
```

```
[143]: arr2[-5,-5]
```

```
[143]: np.int32(49)
```

```
[144]: arr2
```

```
[144]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
             [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
             [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
             [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
             [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
             [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[145]: arr2[-5,-5]
```

```
[145]: np.int32(49)
```

```
[146]: arr2
```

```
[146]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
             [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
             [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
             [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
             [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
```

```
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
[45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
[40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
[98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[147]: arr2[-1,-2]
```

```
[147]: np.int32(20)
```

```
[148]: arr2
```

```
[148]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
[65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
[92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
[60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
[60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
[45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
[40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
[98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[149]: arr2[::-1]
```

```
[149]: array([[98, 53, 68, 50, 59, 78, 61, 79, 20, 51],
[40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
[45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
[63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
[60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
[92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
[65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
[36, 63, 77, 73, 23, 60, 18, 36, 37, 47]], dtype=int32)
```

```
[150]: arr2[::-1]
```

```
[150]: array([[98, 53, 68, 50, 59, 78, 61, 79, 20, 51],
[40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
[45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
[63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
[60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
[92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
[65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
```

```
[36, 63, 77, 73, 23, 60, 18, 36, 37, 47]], dtype=int32)
```

```
[151]: arr2
```

```
[151]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
           [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
           [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
           [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
           [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
           [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
           [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
           [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
           [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
           [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[152]: arr2[::-2]
```

```
[152]: array([[98, 53, 68, 50, 59, 78, 61, 79, 20, 51],
           [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
           [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
           [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
           [65, 60, 68, 46, 49, 99, 68, 13, 35, 93]], dtype=int32)
```

```
[153]: arr2
```

```
[153]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
           [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
           [92, 18,  0, 90, 16, 50, 87, 17, 99,  8],
           [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
           [60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
           [ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
           [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
           [45, 15, 32, 59, 87,  0, 89, 79, 34, 70],
           [40, 65, 55, 61, 28, 41, 47, 65, 94,  4],
           [98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[154]: arr2[::-3]
```

```
[154]: array([[98, 53, 68, 50, 59, 78, 61, 79, 20, 51],
           [63, 23,  3, 31, 80, 80, 80, 10, 71, 80],
           [60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
           [36, 63, 77, 73, 23, 60, 18, 36, 37, 47]], dtype=int32)
```

```
[155]: arr2
```

```
[155]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
           [65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
```

```
[92, 18, 0, 90, 16, 50, 87, 17, 99, 8],
[60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
[60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[63, 23, 3, 31, 80, 80, 80, 10, 71, 80],
[45, 15, 32, 59, 87, 0, 89, 79, 34, 70],
[40, 65, 55, 61, 28, 41, 47, 65, 94, 4],
[98, 53, 68, 50, 59, 78, 61, 79, 20, 51]], dtype=int32)
```

```
[156]: arr2[:-3]
```

```
[156]: array([[36, 63, 77, 73, 23, 60, 18, 36, 37, 47],
[65, 60, 68, 46, 49, 99, 68, 13, 35, 93],
[92, 18, 0, 90, 16, 50, 87, 17, 99, 8],
[60, 56, 41, 16, 73, 75, 93, 36, 10, 32],
[60, 72, 91, 20, 80, 66, 89, 64, 12, 21],
[ 4, 88, 48, 88, 76, 49, 40, 22, 65, 27],
[63, 23, 3, 31, 80, 80, 80, 10, 71, 80]], dtype=int32)
```

```
[157]: arr
```

```
[157]: array([0, 1, 2, 3, 4, 5])
```

```
[158]: arr.max()
```

```
[158]: np.int64(5)
```

```
[159]: arr.min()
```

```
[159]: np.int64(0)
```

```
[160]: arr
```

```
[160]: array([0, 1, 2, 3, 4, 5])
```

```
[161]: arr.mean()
```

```
[161]: np.float64(2.5)
```

```
[162]: arr
```

```
[162]: array([0, 1, 2, 3, 4, 5])
```

```
[163]: arr.median()
```

```
-----
AttributeError
```

```
Traceback (most recent call last)
```

```
Cell In[163], line 1
----> 1 arr.median()
```

```
AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

```
[165]: from numpy import *
a = array([1,2,3,4,9])
median(a)
```

```
[165]: np.float64(3.0)
```

```
[ ]: Without work on import* can you please find the median, mode)
```

```
[167]: arr
```

```
[167]: array([0, 1, 2, 3, 4, 5])
```

```
[168]: arr.reshape(3,2)
```

```
[168]: array([[0, 1],
             [2, 3],
             [4, 5]])
```

```
[169]: arr.reshape(6,1)
```

```
[169]: array([[0],
             [1],
             [2],
             [3],
             [4],
             [5]])
```

```
[170]: arr.reshape(1,6)
```

```
[170]: array([[0, 1, 2, 3, 4, 5]])
```

```
[171]: arr
```

```
[171]: array([0, 1, 2, 3, 4, 5])
```

```
[172]: arr.reshape(2,4)
```

```
-----
ValueError
```

```
Traceback (most recent call last)
```

```
Cell In[172], line 1
----> 1 arr.reshape(2,4)
```

```
ValueError: cannot reshape array of size 6 into shape (2,4)
```

```
[173]: arr
```

```
[173]: array([0, 1, 2, 3, 4, 5])
```

```
[174]: arr.reshape(2,3,order='C')
```

```
[174]: array([[0, 1, 2],  
            [3, 4, 5]])
```

```
[175]: arr.reshape(2,3,order='F') # print element with fortran
```

```
[175]: array([[0, 2, 4],  
            [1, 3, 5]])
```

```
[176]: arr.reshape(2,3,order='A') # A almost give you c type output
```

```
[176]: array([[0, 1, 2],  
            [3, 4, 5]])
```

```
[177]: arr
```

```
[177]: array([0, 1, 2, 3, 4, 5])
```

```
[178]: arr.reshape(2,3)
```

```
[178]: array([[0, 1, 2],  
            [3, 4, 5]])
```

```
[179]: arr.reshape(1,4)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[179], line 1  
----> 1 arr.reshape(1,4)  
  
ValueError: cannot reshape array of size 6 into shape (1,4)
```

```
[180]: arr.reshape(1,6)
```

```
[180]: array([[0, 1, 2, 3, 4, 5]])
```

```
[181]: arr.reshape(6,1)
```

```
[181]: array([[0],
            [1],
            [2],
            [3],
            [4],
            [5]])
```

```
[182]: arr.reshape(2,6)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[182], line 1
----> 1 arr.reshape(2,6)

ValueError: cannot reshape array of size 6 into shape (2,6)
```

```
[183]: arr.reshape(3,3)
```

```
-----
ValueError                                Traceback (most recent call last)
Cell In[183], line 1
----> 1 arr.reshape(3,3)

ValueError: cannot reshape array of size 6 into shape (3,3)
```

```
[184]: arr
```

```
[184]: array([0, 1, 2, 3, 4, 5])
```

```
[185]: arr.reshape(3,2)
```

```
[185]: array([[0, 1],
            [2, 3],
            [4, 5]])
```

```
[186]: ## Indexing
```

```
[187]: mat = np.arange(0,100).reshape(10,10)
      mat
```

```
[187]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
            [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
            [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
            [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
```



```
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[188]: row = 4
      col = 5
```

```
[189]: col
```

```
[189]: 5
```

```
[190]: row
```

```
[190]: 4
```

```
[191]: mat
```

```
[191]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[192]: mat[6] # default it represent to rows
```

```
[192]: array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
[193]: mat
```

```
[193]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[194]: # With Slices
       mat[:,col]
```

```
[194]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
[195]: mat
```

```
[195]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[196]: mat[row,:]
```

```
[196]: array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
[197]: mat
```

```
[197]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[198]: mat
```

```
[198]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[199]: mat[:]
```

```
[199]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[200]: mat[:,8]
```

```
[200]: array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
[201]: mat
```

```
[201]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[202]: mat[:, -1]
```

```
[202]: array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
[203]: mat
```

```
[203]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[204]: row
```

```
[204]: 4
```

```
[205]: col
```

```
[205]: 5
```

```
[206]: mat[:,col]
```

```
[206]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
[207]: mat
```

```
[207]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[208]: mat[1,4]
```

```
[208]: np.int64(14)
```

```
[209]: mat
```

```
[209]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[210]: mat[1:4]
```

```
[210]: array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
[211]: mat[3:-3]
```

```
[211]: array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
            [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
            [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
[212]: mat[0]
```

```
[212]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
[213]: mat[6]
```

```
[213]: array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
[214]: mat
```

```
[214]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
            [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
            [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
            [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
            [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
            [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
            [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
            [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
            [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[215]: mat[6:]
```

```
[215]: array([[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
            [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
            [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
            [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[216]: mat[:6]
```

```
[216]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
            [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
            [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
            [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
            [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
[217]: mat[5:7]
```

```
[217]: array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
[218]: mat[0:10]
```

```
[218]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[219]: mat[0:10:3]
```

```
[219]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[220]: mat[0:10:3]
```

```
[220]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[221]: mat[4:]
```

```
[221]: array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[222]: mat[::-1]
```

```
[222]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
```

```
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
[223]: mat[:, :-3]
```

```
[223]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
[224]: mat[:, :-3]
```

```
[224]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
[225]: mat[:, :-5]
```

```
[225]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
[226]: mat[2:6]
```

```
[226]: array([[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
[227]: mat[2:6, 2:4] # 1:5 --> only row part /// 1:3 -- it indicates only column parts
```

```
[227]: array([[22, 23],
[32, 33],
[42, 43],
[52, 53]])
```

```
[228]: mat[0, 1]
```

```
[228]: np.int64(1)
```

```
[229]: mat[1, 6]
```

```
[229]: np.int64(16)
```

```
[230]: mat[1:]
```

```
[230]: array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[231]: mat
```

```
[231]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
             [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
             [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
             [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
             [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[232]: mat[:6]
```

```
[232]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
             [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
             [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
             [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
             [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
             [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
[233]: mat[0:1]
```

```
[233]: array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]])
```

```
[234]: mat[1:2,2:4]
```

```
[234]: array([[12, 13]])
```

```
[235]: mat[2:3,2:3]
```

```
[235]: array([[22]])
```

```
[236]: mat[3:5,2:4]
```



```
[236]: array([[32, 33],
            [42, 43]])
```

```
[237]: mat[2:3,4:5]
```

```
[237]: array([[24]])
```

```
[ ]: ## Masking
```

```
[238]: mat # we also called as filter
```

```
[238]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
            [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
            [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
            [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
            [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
            [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
            [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
            [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
            [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[239]: id(mat)
```

```
[239]: 2123381819600
```

```
[241]: mat
```

```
[241]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
            [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
            [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
            [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
            [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
            [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
            [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
            [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
            [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
            [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[242]: mat[mat<50]
```

```
[242]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
            17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
            34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
[243]: mat > 50
```

```
[243]: array([[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True],
[ True, True, True, True, True, True, True, True, True,
True]])
```

```
[244]: mat[mat==50]
```

```
[244]: array([50])
```

```
[245]: mat
```

```
[245]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
[246]: mat == 50
```

```
[246]: array([[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False]])
```

```

False],
[False, False, False, False, False, False, False, False, False,
False],
[ True, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False],
[False, False, False, False, False, False, False, False, False,
False]]))

```

```

[247]: a1 = mat[mat<50]
a1

```

```

[247]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
          17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
          34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])

```

```

[248]: a2 = mat[mat>50]
a2

```

```

[248]: array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
          68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
          85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])

```

```

[249]: a3 = mat[mat<=50]
a3

```

```

[249]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
          17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
          34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])

```

```

[250]: a4 = mat[mat==50]
a4

```

```

[250]: array([50])

```

```

[ ]: # python program to generat otp

```

```

[251]: import random

def generate_otp(length=4):
    """Generate a numeric OTP of a specified length."""
    digits = '012345'

```

```

    otp = ''.join(random.choice(digits) for _ in range(length))
    return otp

# Example usage
otp_length = 4 # You can change this to any length you prefer
otp = generate_otp(otp_length)
print(f"Your OTP is: {otp}")

```

Your OTP is: 0325

```

[252]: def wish():
        print('good even')
        wish()

        def wish():
            print('good even')
            wish()

        def wish():
            print('good even')
            wish()

```

good even
good even
good even

```

[253]: list1=['a','b','g',1,5]
        print(list1.pop)

```

<built-in method pop of list object at 0x000001EE6324D380>

```

[254]: x = [1, 2, 3]
        y = x.copy()
        x.append(4)
        print(x)

```

[1, 2, 3, 4]